

Analog Electronic Circuits : Problem Set 5

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For all the problems, use $\mu_n C_{ox}(W/L) = 2 \text{ mA/V}^2$ and $V_T = 0.7 \text{ V}$. Assume that $\lambda = 0$ unless otherwise mentioned. All capacitors are infinite.

Problem 1

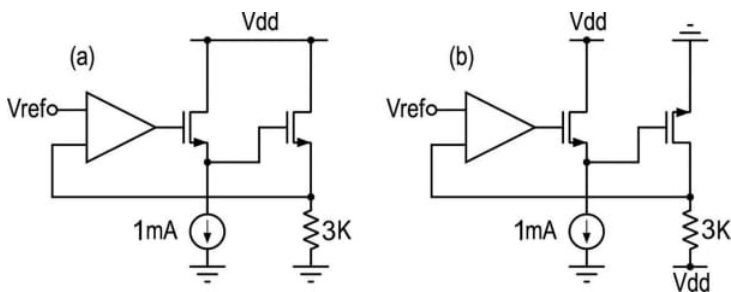


Figure 1: Circuit for problem 1.

In Fig. 1(a), determine the signs on the opamp for negative feedback operation. Determine the gate potentials of the transistors for Vref=3 V and Vdd=7 V. Repeat for Vref=3 V and Vdd=5 V.

In Fig. 1(b), determine the signs on the opamp for negative feedback operation. Determine the gate potentials of the transistors for Vref=3 V and Vdd=6 V. Repeat for Vref=3 V and Vdd=2.2 V.

Problem 2

Fig. 2 shows a progression of current mirrors. For this problem, assume that all transistors have $\lambda = 1/25 \text{ V}$. Further, use Vdd = 10 V. To find the potentials at various nodes of the mirrors, assume that $\lambda = 0$. Use this information to determine the gm's and ro's of the transistors. The opamps in parts (f) and (h) are ideal. Determine the signs on these opamps for negative feedback operation.

For each of the current mirrors, determine

- The minimum voltage Vo needed to ensure that the output behaves like a current source.

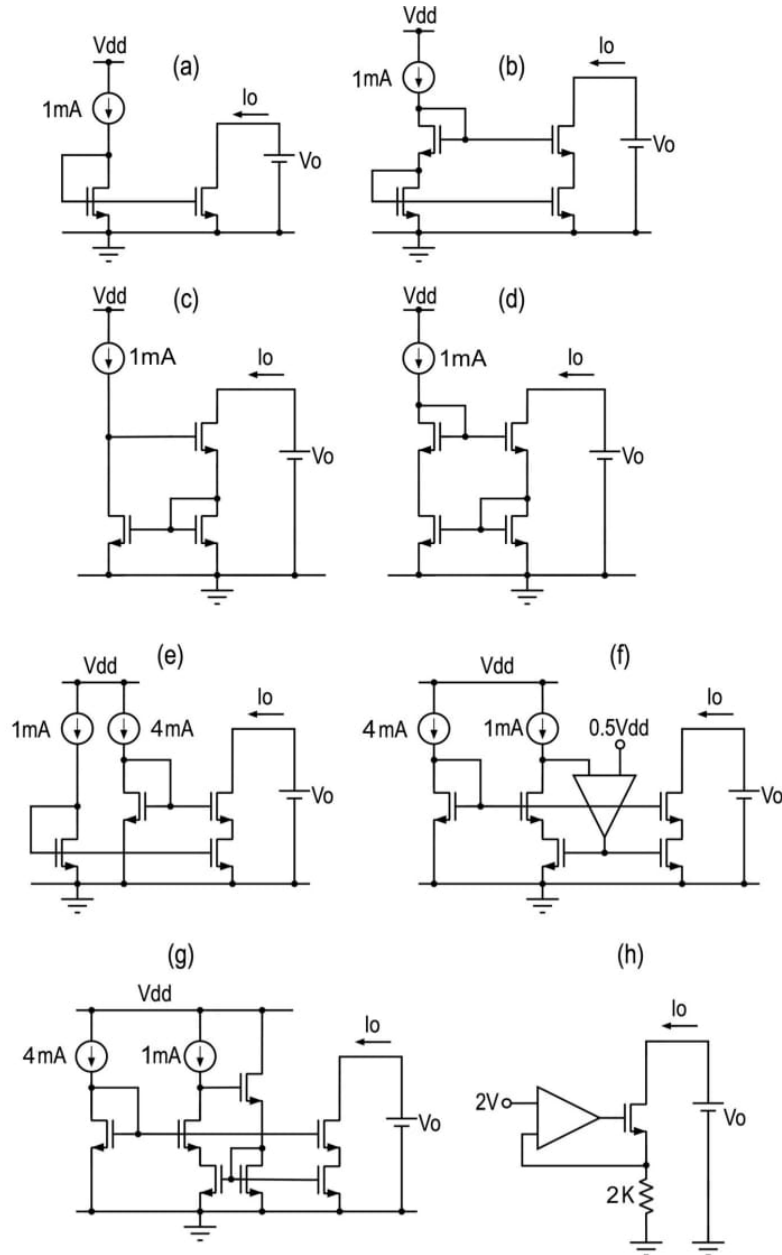


Figure 2: Circuit for problem 2.

- The incremental output resistance, assuming all devices are in saturation.
- Current error, which we will define as $(I_o - 1 \text{ mA})/I_o$ when $V_o = 0.5 V_{dd}$.